# 40V SILICON HIGH CURRENT SCHOTTKY BARRIER DIODE

## Product Summary

- $V_R$ = 40V
- $I_F$ = 2A

## Description and Applications

A surface mount Schottky Barrier Diode featuring low forward voltage drop suitable for high frequency rectification and reverse voltage protection.

- Mobiles
- DC-DC converters
- High frequency rectification

## Features and Benefits

- High Current Capability
- Low Forward Voltage
- Fast Recovery Time
- Small Package Size
- **Totally Lead-Free & Fully RoHS Compliant (Notes 1 & 2)**
- **Halogen and Antimony Free. "Green" Device (Note 3)**
- **For automotive applications requiring specific change control (i.e. parts qualified to AEC-Q100/101/104/200, PPAP capable, and manufactured in IATF 16949 certified facilities), please contact us or your local Diodes representative. https://www.diodes.com/quality/product-definitions/**

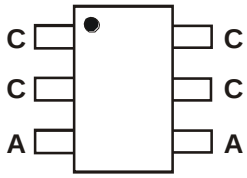
## Mechanical Data

- Package: SOT26
- Package Material: Molded Plastic, "Green" Molding Compound; UL Flammability Classification Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020
- Terminals: Matte Tin Finish Annealed over Copper Leadframe; (Lead-Free Plating) Solderable per MIL-STD-202, Method 208 e3
- Weight: 0.016 grams (Approximate)

SOT26

Top View

Device Symbol



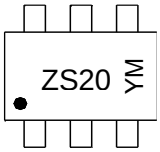

Top View
Pin Out

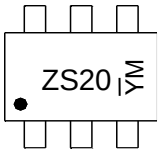
## Ordering Information (Note 4)

| Part Number | Package | Packing | |
|---|---|---|---|
| | | Qty. | Carrier |
| ZHCS2000TA | SOT26 | 3,000 | Tape & Reel |

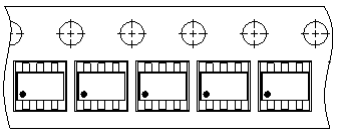
Notes:
1. No purposely added lead. Fully EU Directive 2002/95/EC (RoHS), 2011/65/EU (RoHS 2) & 2015/863/EU (RoHS 3) compliant.
2. See https://www.diodes.com/quality/lead-free/ for more information about Diodes Incorporated's definitions of Halogen- and Antimony-free, "Green" and Lead-free.
3. Halogen- and Antimony-free "Green" products are defined as those which contain <900ppm bromine, <900ppm chlorine (<1500ppm total Br + Cl) and <1000ppm antimony compounds.
4. For packaging details, go to our website at https://www.diodes.com/design/support/packaging/diodes-packaging/.

## Marking Information




ZS20 = Product Type Marking Code
YM = Date Code Marking
Y or $\overline{Y}$ = Year (ex: J = 2022)
M = Month (ex: 9 = September)



Date Code Key

| Year | 2011 | …. | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2031 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Code | Y | …. | J | K | L | M | N | O | P | R | S | T |

| Month | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Code | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | O | N | D |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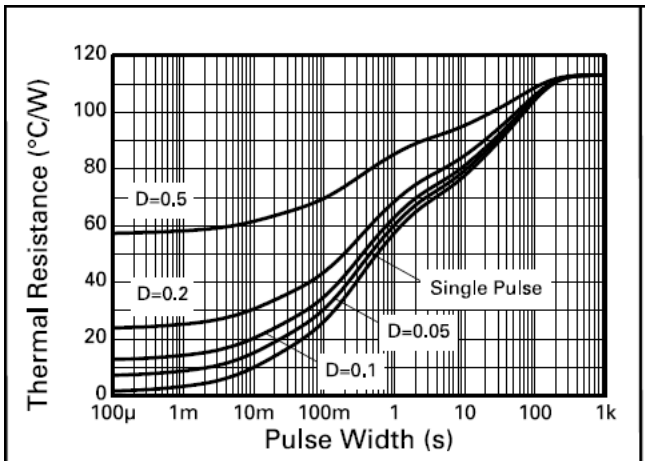
## Maximum Ratings (@$T_A$ = +25°C unless otherwise specified.)

| Characteristic | | Symbol | Value | Units |
|---|---|---|---|---|
| Continuous Reverse Voltage | | $V_R$ | 40 | V |
| Continuous Forward Current | | $I_F$ | 2 | A |
| Average Peak Forward Current; D.C. = 50% | | $I_{FAV}$ | 4 | A |
| Non Repetitive Forward Current | t ≤ 100µs | $I_{FSM}$ | 20 | A |
| | t ≤ 10ms | | 10 | A |

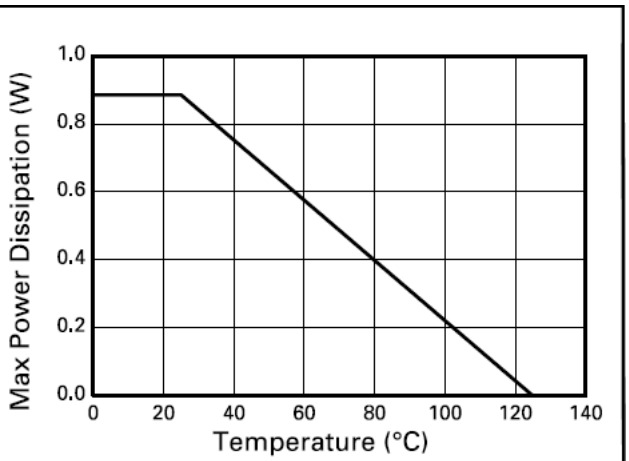
## Thermal Characteristics

| Characteristic | | Symbol | Value | Unit |
|---|---|---|---|---|
| Power Dissipation, $T_A$ = +25°C | | $P_D$ | 1.1 | W |
| Thermal Resistance, Junction to Ambient | (Note 5)<br>(Note 6) | $R_{\theta JA}$ | 113<br>73 | — |
| Junction Temperature | | $T_J$ | +125 | °C |
| Storage Temperature Range | | $T_{STG}$ | -55 to +150 | °C |

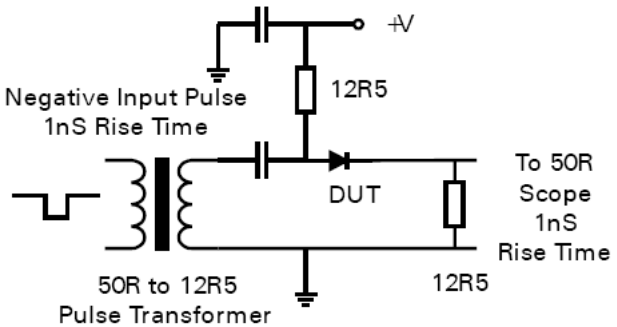
Notes: 5. For a device surface mounted on 25mm x 25mm FR4 PCB with high coverage of single sided 1oz copper, in still air conditions.
6. For a device surface mounted on FR4 PCB measured at t ≤ 5 secs.



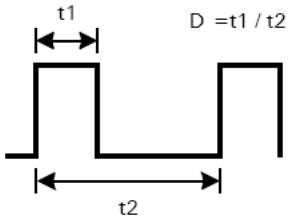

**Transient Thermal Impedance**




**Derating Curve**




**Reverse Recovery Time Circuit**




**Duty Cycle**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## Electrical Characteristics (@$T_A$ = +25°C, unless otherwise specified.)

| Characteristic | Symbol | Min | Typ | Max | Unit | Test Condition |
|---|---|---|---|---|---|---|
| Reverse Breakdown Voltage | $V_{(BR)R}$ | 40 | — | — | V | $I_R$ = 1mA |
| Forward Voltage (Note 7) | $V_F$ | — | 290 | 325 | mV | $I_F$ = 500mA |
| | | — | 340 | 385 | | $I_F$ = 1000mA |
| | | — | 380 | 445 | | $I_F$ = 1500mA |
| | | — | 420 | 500 | | $I_F$ = 2000mA |
| | | — | 485 | 615 | | $I_F$ = 3000mA |
| | | — | 420 | - | | $I_F$ = 2000mA, $T_A$ = +100°C |
| Reverse Current | $I_R$ | — | 160 | 300 | µA | $V_R$ = 30V |
| Diode Capacitance | $C_D$ | — | 50 | — | pF | f = 1MHz, $V_R$ = 25V |
| Reverse Recovery Time | $t_{rr}$ | — | 5.5 | — | ns | Switched from $I_F$ = 500mA to $I_R$ = 500mA Measured @ $I_R$ = 50mA |

Note: 7. Measured under pulsed conditions. Pulse width = 300µs. Duty cycle ≤ 2%.


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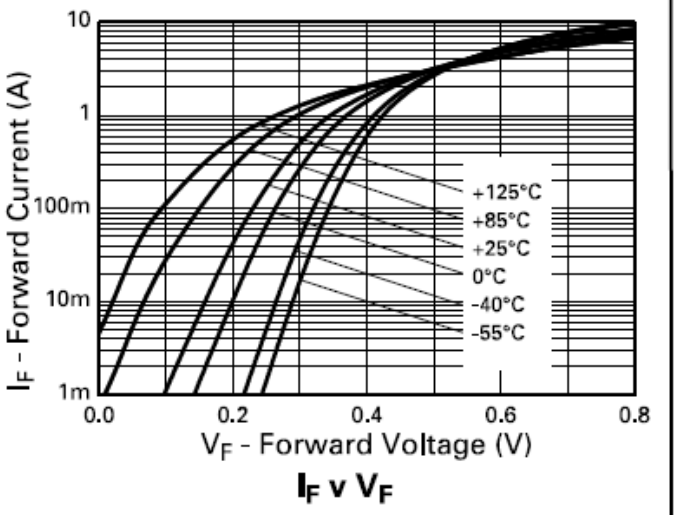
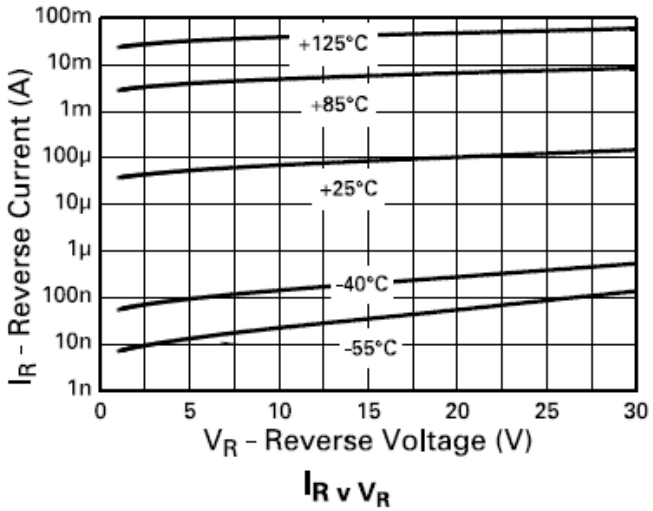
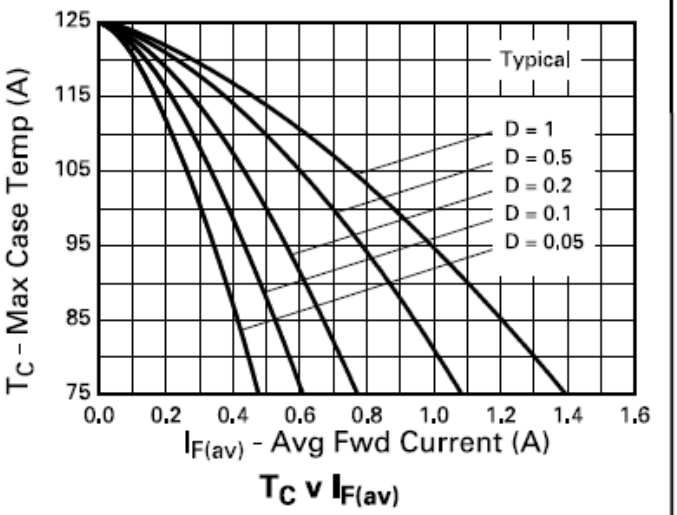


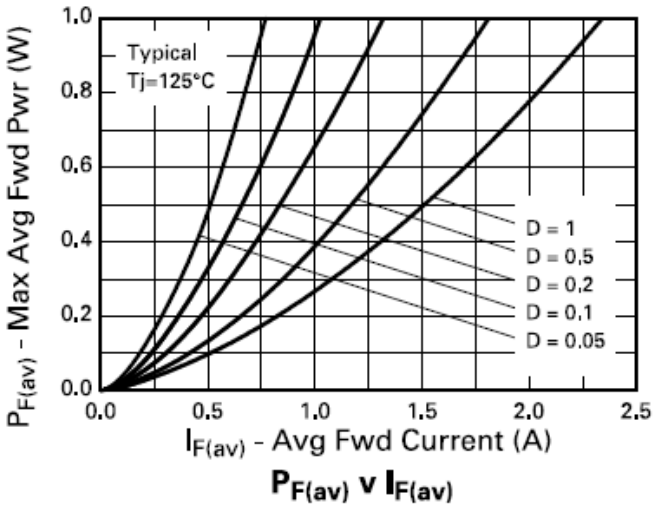

**$I_F$ v $V_F$**



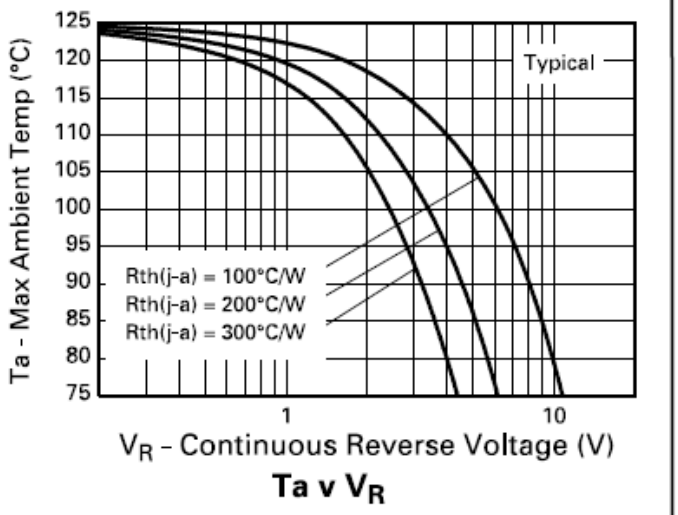

**$I_R$ v $V_R$**



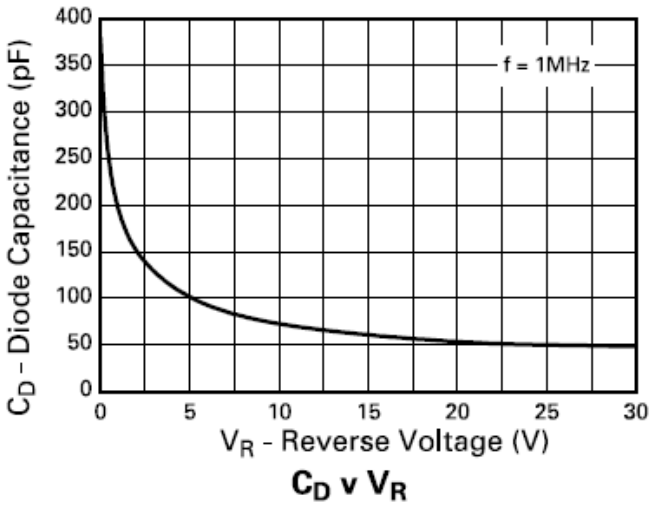

**$T_C$ v $I_{F(av)}$**




**$P_{F(av)}$ v $I_{F(av)}$**




**Ta v $V_R$**




**$C_D$ v $V_R$**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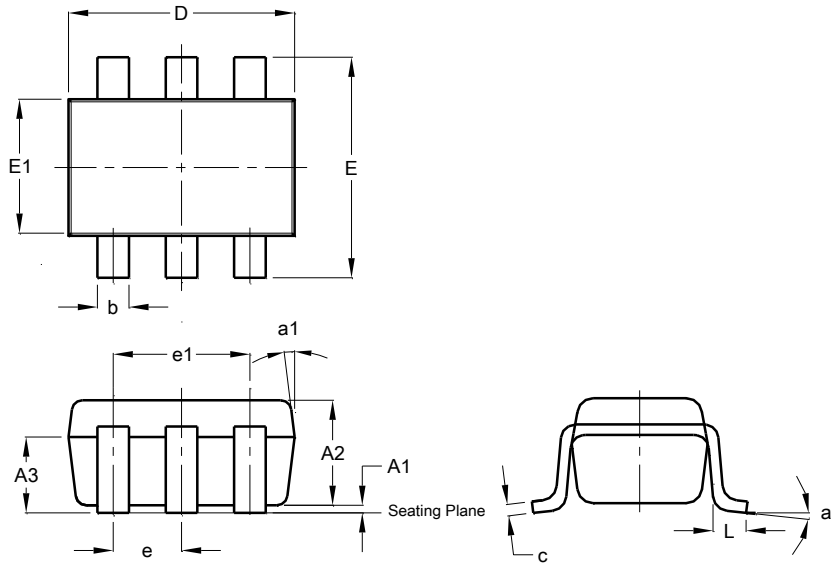
## Package Outline Dimensions

Please see http://www.diodes.com/package-outlines.html for the latest version.

**SOT26**

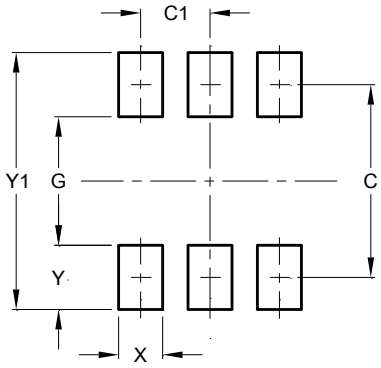



| SOT26 | | | |
|---|---|---|---|
| **Dim** | **Min** | **Max** | **Typ** |
| **A1** | 0.013 | 0.10 | 0.05 |
| **A2** | 1.00 | 1.30 | 1.10 |
| **A3** | 0.70 | 0.80 | 0.75 |
| **b** | 0.35 | 0.50 | 0.38 |
| **c** | 0.10 | 0.20 | 0.15 |
| **D** | 2.90 | 3.10 | 3.00 |
| **e** | - | - | 0.95 |
| **e1** | - | - | 1.90 |
| **E** | 2.70 | 3.00 | 2.80 |
| **E1** | 1.50 | 1.70 | 1.60 |
| **L** | 0.35 | 0.55 | 0.40 |
| **a** | - | - | 8° |
| **a1** | - | - | 7° |
| **All Dimensions in mm** | | | |

## Suggested Pad Layout

Please see http://www.diodes.com/package-outlines.html for the latest version.

**SOT26**




| Dimensions | Value (in mm) |
|---|---|
| **C** | 2.40 |
| **C1** | 0.95 |
| **G** | 1.60 |
| **X** | 0.55 |
| **Y** | 0.80 |
| **Y1** | 3.20 |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